

SECURITY INCIDENT INVESTIGATION REPORT

Windows Sysmon Security Incident Analysis

Analyst

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Role

Future SOC Analyst | Future Cloud Security Engineer

Investigation Date

June 2026

Classification

Internal Security Monitoring Investigation

Severity

Low

Status

Closed

1. Executive Summary

A Windows endpoint monitoring investigation was conducted using Microsoft Sysmon and Windows Event Viewer to assess system activity and identify potential indicators of compromise (IOCs).

The investigation focused on endpoint telemetry generated through process creation events, network connections, file creation activities, image loading operations, DNS queries, and registry modifications. The objective was to validate Sysmon logging capabilities, perform threat hunting activities, and determine whether suspicious or malicious activity was present on the monitored endpoint.

During the investigation, Sysmon successfully captured multiple security-relevant events. Process execution, browser network communications, file system changes, DLL loading activity, DNS requests, and registry modifications were reviewed and correlated. Analysis determined that all observed events were consistent with legitimate operating system and user-generated activity.

No evidence of malware execution, persistence mechanisms, privilege escalation attempts, credential theft, unauthorized remote access, or command-and-control communication was identified.

The investigation concluded with a LOW risk assessment and no indicators of compromise detected.

2. Incident Overview

As part of a proactive security monitoring exercise, endpoint telemetry was collected from a Windows 11 workstation using Microsoft Sysmon.

The objective of the investigation was to validate Sysmon event collection, identify security-relevant activities, and assess the endpoint for suspicious behaviors commonly associated with adversary techniques.

The following Sysmon event categories were selected for investigation:

- Event ID 1 – Process Creation
- Event ID 3 – Network Connections
- Event ID 7 – Image Load
- Event ID 11 – File Creation
- Event ID 13 – Registry Value Set
- Event ID 22 – DNS Query

The investigation was performed using Windows Event Viewer and exported Sysmon logs (.evtx) for evidence preservation and analysis.

3. Investigation Methodology

The investigation followed a structured SOC analyst workflow:

1. Verify Sysmon deployment and event generation
 2. Identify high-value Sysmon event categories
 3. Review process execution telemetry
 4. Analyze network communication events
 5. Investigate file creation activities
 6. Validate DLL and module loading operations
 7. Review DNS query telemetry
 8. Examine registry modification events
 9. Correlate findings
 10. Map observed activity to MITRE ATT&CK
 11. Document findings and recommendations
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4. Process Creation Analysis

Event ID 1 – Process Creation

Observed Activity

The investigation identified execution of the following processes:

- cmd.exe
- ipconfig.exe
- rundll32.exe

Evidence

Process:

ipconfig.exe

Parent Process:

cmd.exe

User:

Vasantha\nitin

Analysis

Sysmon successfully captured command-line execution activity initiated by the authenticated user.

The parent-child relationship between cmd.exe and ipconfig.exe was verified and determined to be legitimate.

No suspicious execution patterns were identified.

Specifically, the investigation found no evidence of:

- Encoded PowerShell commands
- Malicious script execution
- LOLBins abuse
- Unauthorized process injection

MITRE ATT&CK Mapping

T1059 – Command and Scripting Interpreter

T1016 – System Network Configuration Discovery

T1033 – System Owner/User Discovery

Risk

Low

5. Network Connection Analysis

Event ID 3 – Network Connection

Observed Activity

Application:

Brave Browser

Protocol:

UDP

Destination Port:

5353

Analysis

Network telemetry revealed browser-generated communication utilizing UDP port 5353.

The traffic was identified as Multicast DNS (mDNS) communication used for local service discovery.

The investigation found no evidence of:

- Suspicious external communication
- Known malicious IP addresses
- Command-and-control infrastructure
- Data exfiltration attempts

MITRE ATT&CK Mapping

T1071 – Application Layer Protocol

Risk

Low

6. File Creation Analysis

Event ID 11 – File Creation

Observed Activity

File Created:

New Text Document.txt

Process:

Explorer.exe

User:

Vasantha\nitin

Analysis

The file creation event was generated through normal user interaction with Windows Explorer.

The activity demonstrated Sysmon's capability to monitor file system modifications and track file creation operations.

No suspicious file creation behavior was observed.

MITRE ATT&CK Mapping

T1074 – Data Staged

Risk

Low

7. Image Load Analysis

Event ID 7 – Image Load

Observed Activity

Process:

taskhostw.exe

Module Loaded:

urlmon.dll

Publisher:

Microsoft Corporation

Signature Status:

Valid

Analysis

The loaded module was verified as a legitimate Microsoft-signed DLL.

The module originated from a trusted system location and exhibited no indicators of DLL hijacking or malicious injection.

MITRE ATT&CK Mapping

T1574 – Hijack Execution Flow

Risk

Low

8. Additional Telemetry Analysis

Event ID 13 – Registry Value Set

Observed registry modifications were determined to be legitimate Windows operating system activity.

MITRE ATT&CK:

T1112 – Modify Registry

Risk:

Low

Event ID 22 – DNS Query

Observed DNS requests originated from standard browser activity.

No suspicious domains, DNS tunneling, or malicious indicators were identified.

MITRE ATT&CK:

T1071.004 – DNS

Risk:

Low

9. MITRE ATT&CK Mapping Summary

Event ID	Technique ID	Technique
1	T1059	Command and Scripting Interpreter
1	T1016	System Network Configuration Discovery
1	T1033	System Owner/User Discovery
3	T1071	Application Layer Protocol
7	T1574	Hijack Execution Flow
11	T1074	Data Staged
13	T1112	Modify Registry
22	T1071.004	DNS

10. Security Findings

Finding	Severity
Process Creation Activity	Informational
Network Communication Activity	Informational
File Creation Events	Informational
Image Load Events	Informational
Registry Modifications	Informational
DNS Query Activity	Informational

No Indicators of Compromise (IOCs) were identified during the investigation.

11. Recommendations

Endpoint Monitoring

- Deploy Sysmon across all Windows endpoints
- Maintain updated Sysmon configurations
- Monitor Event IDs 1, 3, 7, 11, 13, and 22

Detection Engineering

Develop detection rules for:

- PowerShell abuse
- Suspicious command-line execution
- DLL hijacking attempts
- Registry persistence techniques
- DNS-based command and control

SIEM Integration

Forward Sysmon telemetry into:

- Microsoft Sentinel
- Splunk
- Elastic Security
- IBM QRadar

Threat Hunting

Conduct periodic reviews of:

- Process execution activity
 - Network communication patterns
 - Registry modifications
 - DNS activity
 - File creation events
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12. Conclusion

The Windows Sysmon Security Incident Analysis successfully demonstrated the use of endpoint telemetry for security monitoring, threat hunting, and incident investigation.

Through detailed examination of process creation events, network connections, file creation activities, image load operations, DNS queries, and registry modifications, comprehensive visibility into endpoint behavior was achieved.

No malicious activity or indicators of compromise were identified during the investigation. The endpoint exhibited expected operating system and user-generated behavior.

This investigation demonstrates practical SOC analyst competencies including log analysis, event correlation, threat hunting, MITRE ATT&CK mapping, security monitoring, and professional incident reporting. These skills align closely with the responsibilities of Security Operations Center analysts responsible for detecting, investigating, and responding to security events in enterprise environments.

Final Assessment

Investigation Status: CLOSED

Risk Rating: LOW

Indicators of Compromise: NONE IDENTIFIED